

INTRODUCTION AND MOTIVATION

- Hot Jupiters are gas giant exoplanets with masses and radii comparable to Jupiter, orbiting extremely close to their host stars.
- Tidal interactions in these systems transfer orbital energy into the stellar interior, causing the orbit to gradually shrink over time, leading to planetary engulfment.
- Measurements of orbital decay in hot Jupiter systems provide one of the only observational probes of tidal dissipation within stars, quantitatively characterised by the stellar tidal quality factor Q'_* [1].

$$Q'_* = -\frac{27\pi}{2\dot{P}} \left(\frac{M_p}{M_*} \right) \left(\frac{R_*}{a} \right)^5 \left(\frac{\omega_*}{n} - 1 \right)$$

- Understanding the efficiency of tidal dissipation is essential for explaining the long-term evolution and fate of close-in planetary systems.
- However, measurement of Q'_* proves difficult as the orbital decay signal can be produced by alternative mechanisms such as apsidal precession or the Late Travel-Time Effect (LTTE).

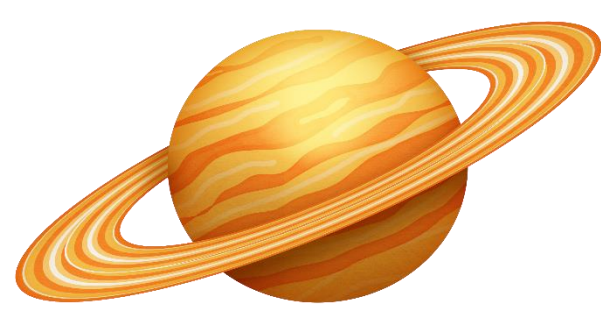
Aims: To test whether the hot Jupiter Kepler-1658b is undergoing genuine orbital decay by comparing it to WASP-12b, a system for which tidal orbital decay is now well-established.

TARGETS – Two Systems, Two Stories

WASP-12b

$$P = 1.09d, M_p = 1.47 \pm 0.076 M_J, R_p = 1.900 \pm 0.057 R_J$$

Benchmark system with confirmed orbital decay



Kepler-1658b

$$P = 3.85d, M_p = 5.88 \pm 0.47 M_J, R_p = 1.07 \pm 0.05 R_J$$

Is the orbit decaying? Is it a planet? Is it a star?

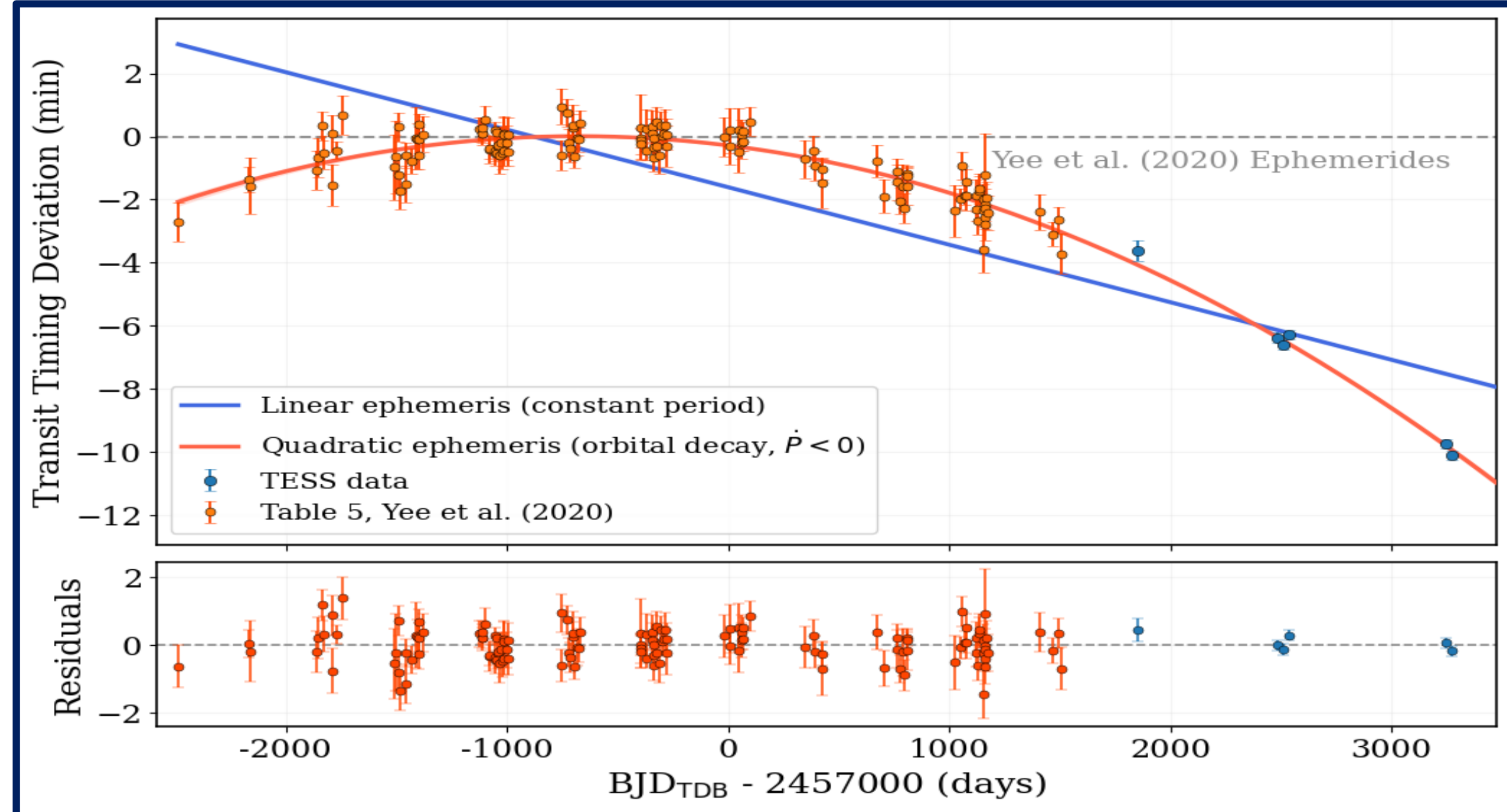


MAIN RESULTS

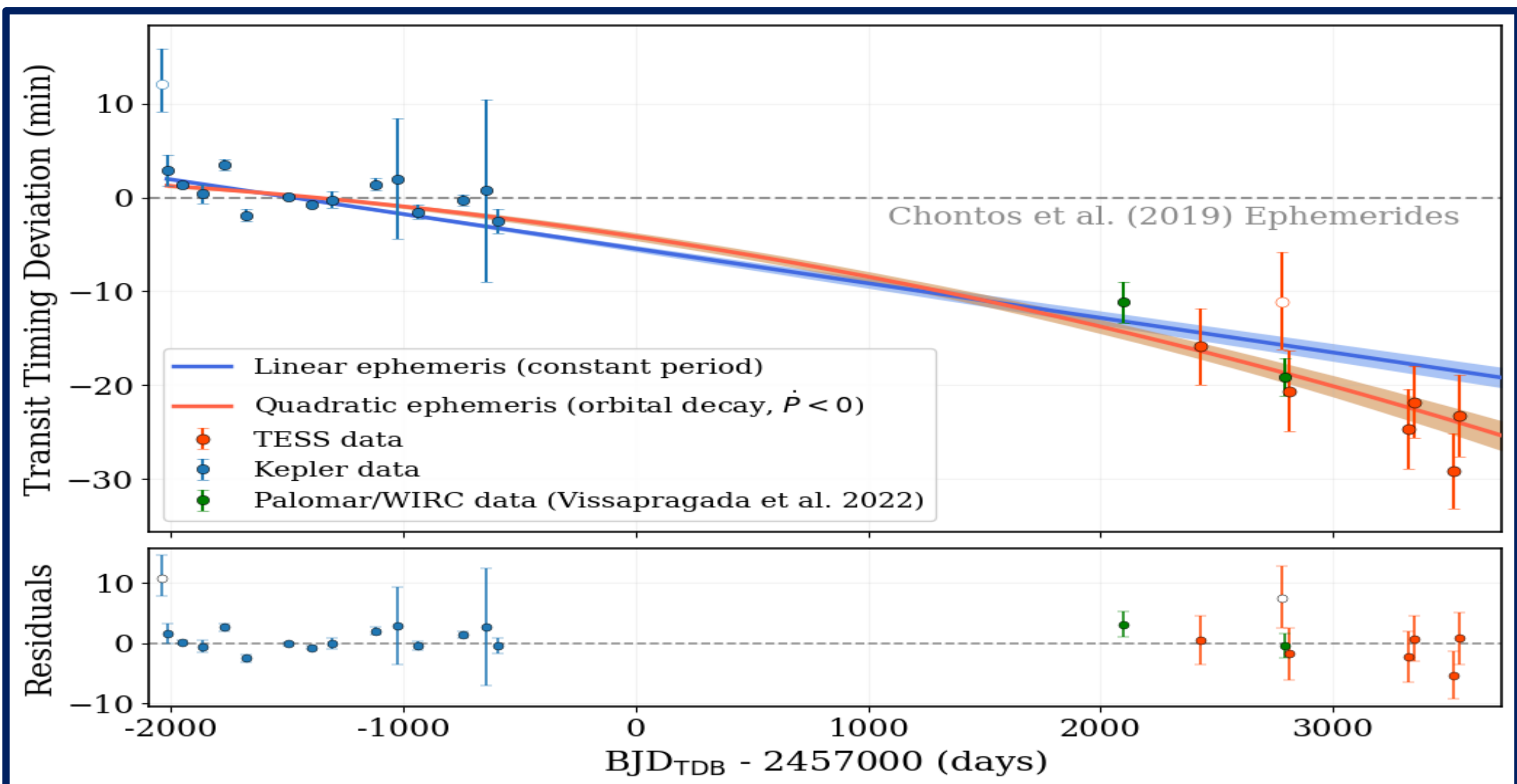
Mid-transit times, t_0 , extracted from the fitted models were used to construct O-C diagrams and test for period evolution. Linear and quadratic ephemerides were fitted to the timing residuals, with the quadratic model describing a uniformly changing orbital period.

$$t(N) = t_0 + NP + \frac{1}{2} \frac{dP}{dN} N^2$$

The quadratic coefficient dP/dN yields $\dot{P}$, from which $\tau = \frac{P}{|\dot{P}|}$ and Q'_* are derived.



O-C diagram of WASP-12b exhibits a clear quadratic trend extending across more than a decade of observations. The newly analysed TESS observations further corroborate previous conclusions in favour of orbital decay.



For Kepler-1658b, the original Kepler observations alone show little evidence for period evolution, but the inclusion of recent TESS observations statistically prefers the quadratic trend. While both systems favour a quadratic ephemeris, the timing signal in WASP-12b is significantly better described by a simple orbital decay model than that of Kepler-1658b.

TRANSIT FITTING

- Planetary orbital decay can be measured using the transit method, which involves detecting the periodic decrease in stellar brightness as a planet passes in front of its host star.
- A changing orbital period causes observed transit times to progressively diverge from those predicted by a constant-period ephemeris, which produces a characteristic trend in an observed-minus-calculated (O-C) diagram.
- The key parameter is the mid-transit time t_0 . Repeated transit observations allow precise measurements of t_0 , making transit photometry a powerful tool for detecting long-term orbital evolution.
- Photometric observations from the Kepler and TESS space telescopes were obtained from the MAST archive and preprocessed using the **Lightcurve** package.
- Transit events were first masked using literature ephemerides and then detrended using the **Savitzky-Golay filter** to remove long-term instrumental and stellar variability.
- Juliet** framework was used for transit modelling via MCMC sampling.
- Global phase-folded light curves were produced by combining all available transits and folding them on the best-fitting orbital period.

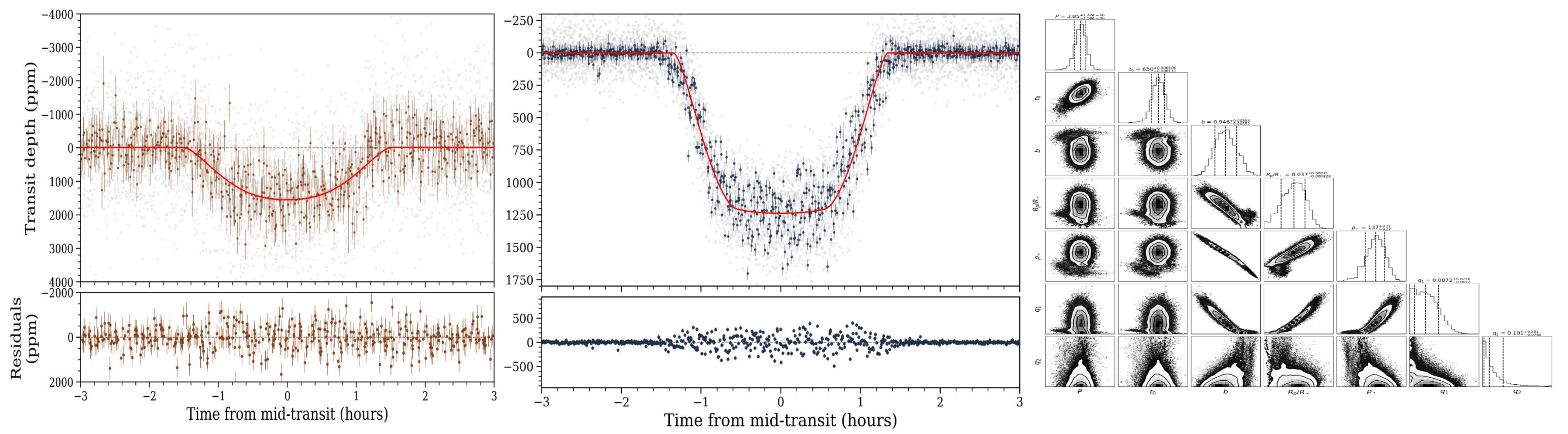
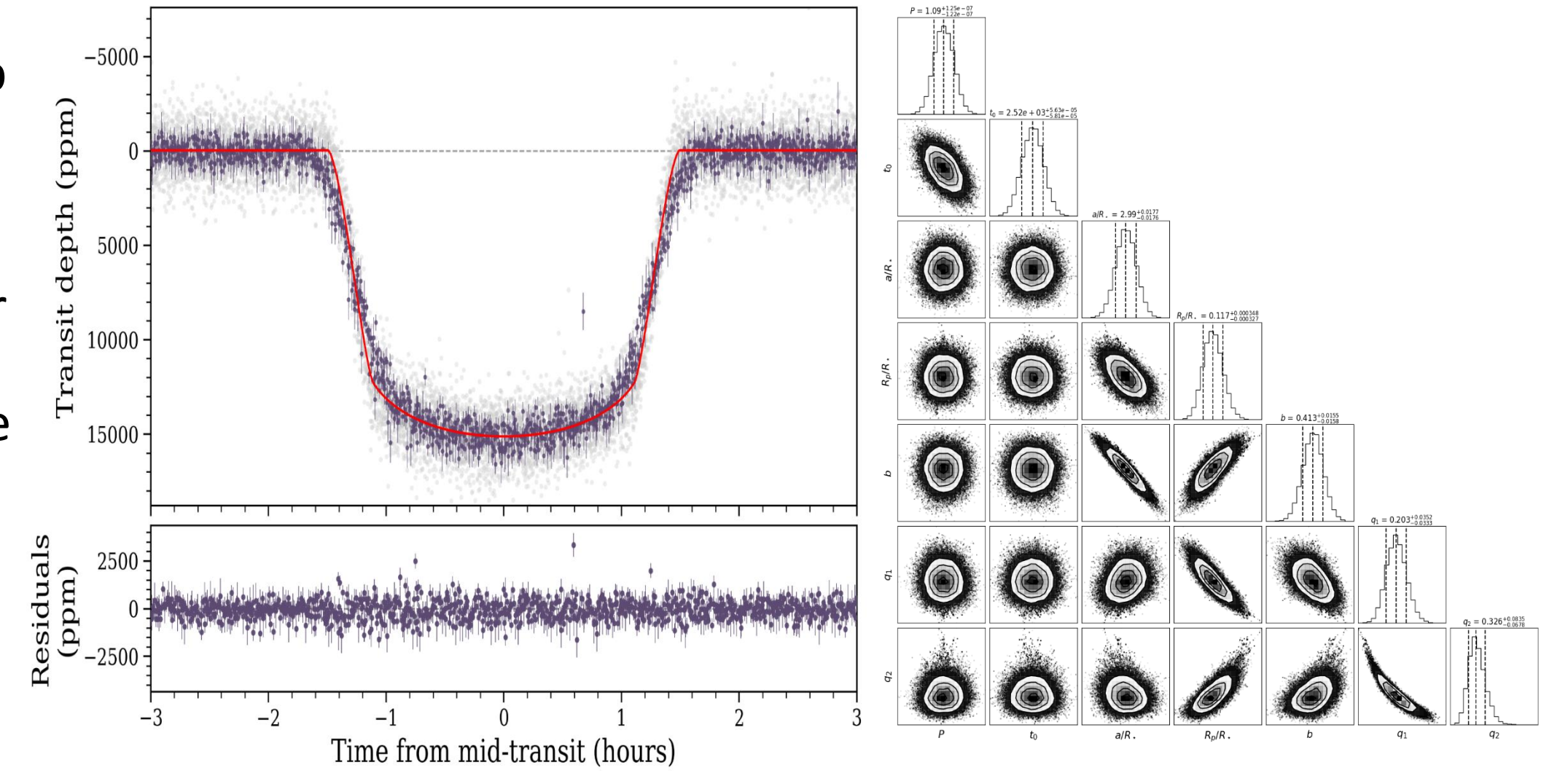


Fig. 1 Transit fitting for Kepler-1658b using lightcurves obtained from TESS (left) and Kepler (middle). The corner plot (right) highlights the strong parameter degeneracies associated with the near-grazing transit geometry of the system.

Fig. 2 Transit fitting for WASP-12b using lightcurves obtained from TESS (left). The corner plot (right) shows the compact, and approximately Gaussian posterior distribution, indicating a well-constrained transit geometry. The low impact parameter compared to Kepler-1658b reduces geometric degeneracies in the fitted parameters.



PARAMETERS	WASP-12b	Kepler-1658b
P (days)	$U[1.08; 1.1]$	$U[3.35; 3.95]$
b	$U[0; 1.11785]$	$U[0; 1]$
R_p/R_*	$N(0.11785; 0.00054)$	$U[0.02; 0.06]$
ρ_* (kg m^{-3})	-	$N(83.4; 7.9)$
a/R_*	$N(3.039; 0.034)$	-
$q1, q2$	$U[0; 1]$	$U[0; 1]$

Table 1 shows the priors used for the transit fitting for Kepler-1658b and WASP-12b. The priors are taken from Yee et al. (2020) [2] for WASP-12b and Chontos et al. (2019) [3] for Kepler-1658b.

ANALYSIS

Derived Parameters	WASP-12b	Wong et al. 2022 [4]	Kepler-1658b	Vissapragada et al. 2022 [5]
$\dot{P}$ (ms/yr)	-30.3 ± 0.9	-29.81 ± 0.94	-87.9 ± 16.8	-131 ± 20
τ (Myr)	3.11 ± 0.09	3.16 ± 0.10	3.79 ± 0.7	≈ 2.5
Q'_*	$(1.82 \pm 0.18) \times 10^5$	$(1.50 \pm 0.11) \times 10^5$	$(1.84 \pm 0.5) \times 10^4$	$(2.50 \pm 0.7) \times 10^4$
ΔBIC	1235.5	—	23.4	—
χ^2	1.12	—	3.50	—

The measured $\dot{P}$, τ and Q'_* for WASP-12b are all in excellent agreement with previous studies, reinforcing its status as the strongest known example of confirmed tidal orbital decay. The interpretation of Kepler-1658b is less straightforward. Although the quadratic ephemeris is statistically preferred, the fit quality remains significantly poorer than for WASP-12b. The inferred Q'_* is also unusually low compared with theoretical expectations for evolved stars. Furthermore, the stellar density ρ_* inferred from transit fitting and the independently measured value through asteroseismology are discrepant by $\sim 4.5\sigma$, suggesting that the simple transiting-planet model may not fully describe the system. These results are consistent with recent studies (e.g. Winn & Stefansson (2025) [6]) proposing that the observed timing variations may arise from LTTE induced by an unresolved companion at ~ 50 au rather than genuine tidal inspiral, and that the system may be explained by a triple star model.

CONCLUSIONS AND FURTHER WORK

Tidal orbital decay is robustly confirmed in WASP-12b. In Kepler-1658b, new TESS sectors drive a statistically significant quadratic trend, but a poor fit, 4.5σ density discrepancy, and anomalously low Q'_* disfavour tidal decay. LTTE from an unresolved companion at ~ 50 au is the more physically consistent interpretation.

A quadratic O-C signal alone is insufficient to confirm orbital decay. Independent tests are essential to rule out alternative mechanisms. Joint transit timing and radial velocity analyses are required to distinguish orbital decay from LTTE. Priority next steps include joint RV and timing analysis of Kepler-1658b with NEID, whereas JWST secondary eclipse timings may definitively distinguish tidal decay from apsidal precession.

REFERENCES

- [1] P. Goldreich and S. Soter, Icarus, 5, 375 (1966)
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- [6] J. N. Winn and G. Stefansson, (2025), arXiv:2510.05229